MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 10AKP7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection and focus, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0 (Note 1)	770	11,000	±125	Note 1	60,000
Minimum:	5.7	-180	---	7,000	---	---	---
TEST CONDITIONS:	6.3	Adjust	650	9,000	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(B) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	---	15	μAdc
5216	Cathode illumination	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Ck to all Cg1 to all Cg2 to all	Ck Cg1 Cg2	--- --- ---	8.0 10.0 10.0	pF pF pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	2.0	mm
1002	Barometric pressure, reduced	3	54 mmHg; maximum voltage applied	---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5208	Gas ratio	6		Gr	---	0.25	---
5101	Neck straightness	4		---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
(B) 5223	Modulation	-	Ib = 200 μAdc	ΔEc1	---	38	Vdc
5231	Spot position (magnetic deflection)	-		---	---	18	mm
5236	Zero-bias anode current	-		---	---	---	---
5241	Grid-cutoff voltage	-		Ec1	-40	-70	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
---	Aperture alinement	2		Distance	---	8.0	mm
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5101	Side terminal and base alinement	-	Pin No. 3	---	---	---	---
5101	Neck and bulb alinement (magnetic types)	-		---	---	---	---
5101	Face tilt	-		---	---	---	---
5216	Stray light emission	-	Eb = 11,000 Vdc; Ec2 = 770 Vdc	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5221	Screens	5		---	---	---	---
5226	Line width A (magnetic deflection)	-	Ib = 200 μAdc	Width	---	0.30	mm
5226	Line width C (magnetic deflection)	-	Ib = 200 μAdc	Width	---	0.44	mm
---	Astigmatism	7		Ratio	---	2	---
5246	Focusing ampere turns	-	Ib = 200 μAdc; D = 3-1/4 inch (82.55 mm)	AT	450	570	---
5251	Heater cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb = 11,000 Vdc; Ec2 = 770 Vdc; Ib = 60 μAdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
5226	Line width A	-		Width	---	0.38	mm
5226	Line width C	-		Width	---	0.55	mm
5223	Modulation	-		ΔEc1	---	38	Vdc

NOTES:

1. When Ec2 is greater than 330 Vdc or Ec1 is near zero, or both, the effective resistance of the anode and grid No. 3 supply should be adequate to limit the anode and grid No. 3 input power to 6 watts.
2. The distance between the center of the unfocused, undeflected spot low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 should not be held at zero for more than approximately 10 seconds to prevent damage to the screen.
3. The test is made with maximum voltages applied to all electrodes connected through the base. Connections should be made in a manner that does not degrade the tubes electrical voltage breakdown characteristics. Criteria for acceptance is the absence of arc-over or corona.
4. Neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5-inches (127.0 mm) long and $1.500\ ^{+0.003}_{-.000}$ inch ($38.1\ ^{+0.08}_{-0.00}$ mm) inside diameter. This cylinder should move freely between the reference line and the base of the assembled tube.

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amendment, revision or specification are the subject of NAT STD A013. When an

NOTES: - Continued

5. The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 400 V minimum and beam current 60 μA maximum. The screen characteristics shall comply with the following minimum limits: cb5=400cb; G5: 1=4.

6. This test to be performed at the conclusion of the holding period.

7. With a single line of 1/60 second duration, and length 9 inches (228.6 mm), modulate grid No. 1 with 1 μs pulses. Adjust peak beam current to 200 μA and focus for the most circular spot at center of screen. Determine orientation for maximum and minimum spot diameters by rotating a partially merged raster, or by an equivalent method. Maintaining focus, determine line widths at these orientations in accordance with procedures described in method 5226. Astigmatism (the ratio of maximum to minimum line width) shall not exceed the value specified.

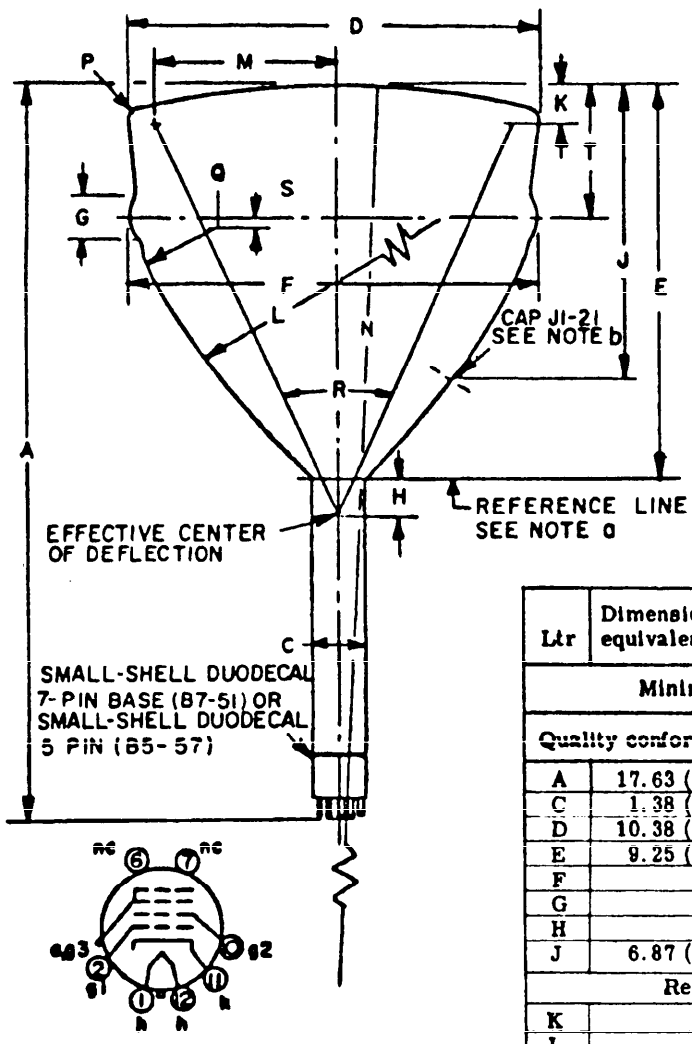
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	17.63 (447.80)	18.00 (457.20)	
C	1.38 (35.05)	1.50 (38.10)	
D	10.38 (263.65)	10.62 (269.75)	
E	9.25 (234.95)	9.62 (244.35)	
F		10.62 (269.75)	
G		.75 (19.05)	
H		.950 (24.13)	
J	6.87 (174.50)	7.62 (193.55)	
Reference dimensions			c
K	.65 (16.51)		
L	20.00 (508.00) RAD		
M	4.50 (114.30) RAD MIN useful screen		
N	42.00 (1066.80) RAD		
P	.38 (9.65)		
Q	2.00 (50.80) RAD		
R	50°		
S	.242 (6.15)		
T	2.90 (73.66)		

Notes
a. Reference line determined by position where JEDEC gage No. 112 will rest on bulb cone.
b. The plane through the tube axis and vacant pin position No. 3 may vary from the plane through the tube axis and anode terminal by an angular tolerance (measured about the tube axis) of 10°. Anode terminal is on the same side as vacant pin position No. 3.
c. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 10AKP7.

5677-70

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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